DATA ANALYSIS PROJECT · DIGITERA TRAINING

PharmaPulse Analytics

Sales, Loss & Inventory Intelligence for Pharmacy Chains

End to end: Python · SQL Server · Power BI · DAX

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The business problem

A pharmacy chain with 12 stores wants to know where profit leaks, which discounts pay off, and where stock is at risk.

Raw data
3 sales files + 8 tables

Python + Power Query
Clean and merge

SQL Server
Explore and validate

Power BI
Model, DAX, dashboard

Insights
Decisions

Python: cleaning the data

```
# two date formats in the raw files
is_slash = d.str.match(r"^\d{2}/\d{2}/\d{4}$")
parsed[is_slash]  = to_datetime(d[is_slash],  format="%d/%m/%Y")
parsed[~is_slash] = to_datetime(d[~is_slash], format="%Y-%m-%d")
# flag the sales that lose money
sales["is_loss"] = (sales["profit"] < 0).astype(int)
```

What the script handles

- Merges 3 sales files
- Removes 50 duplicate rows and 42 duplicate customers
- Extracts promo codes from free-text notes
- Adds profit, margin, is_loss, discount_band

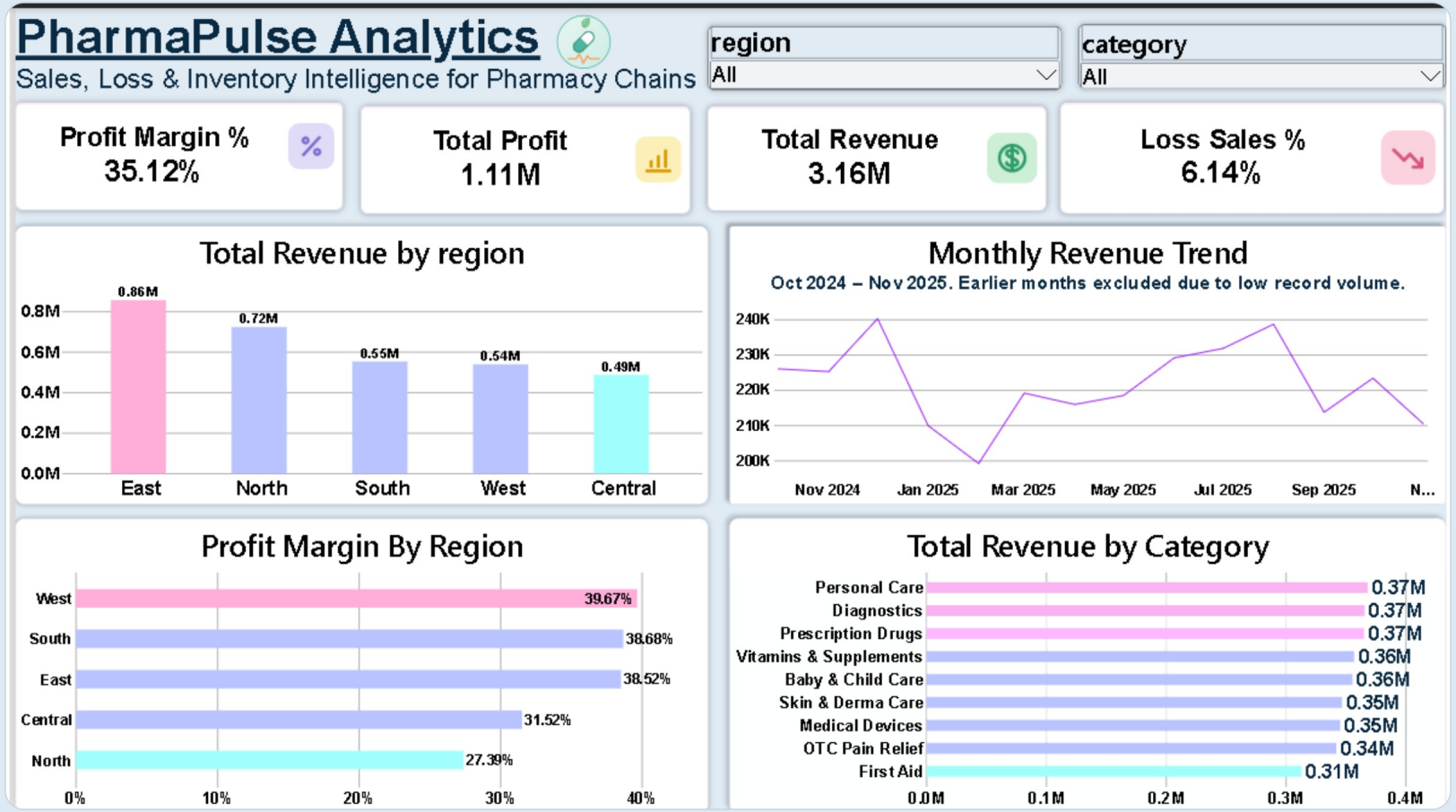
SQL: do bigger discounts hurt margin?

```
SELECT discount_band,
  ROUND(100.0*SUM(profit)/SUM(total_amount),2) AS margin_pct,
  ROUND(100.0*SUM(is_loss)/COUNT(*),2) AS loss_rate_pct
FROM dbo.Sales_cleaned
GROUP BY discount_band;
```

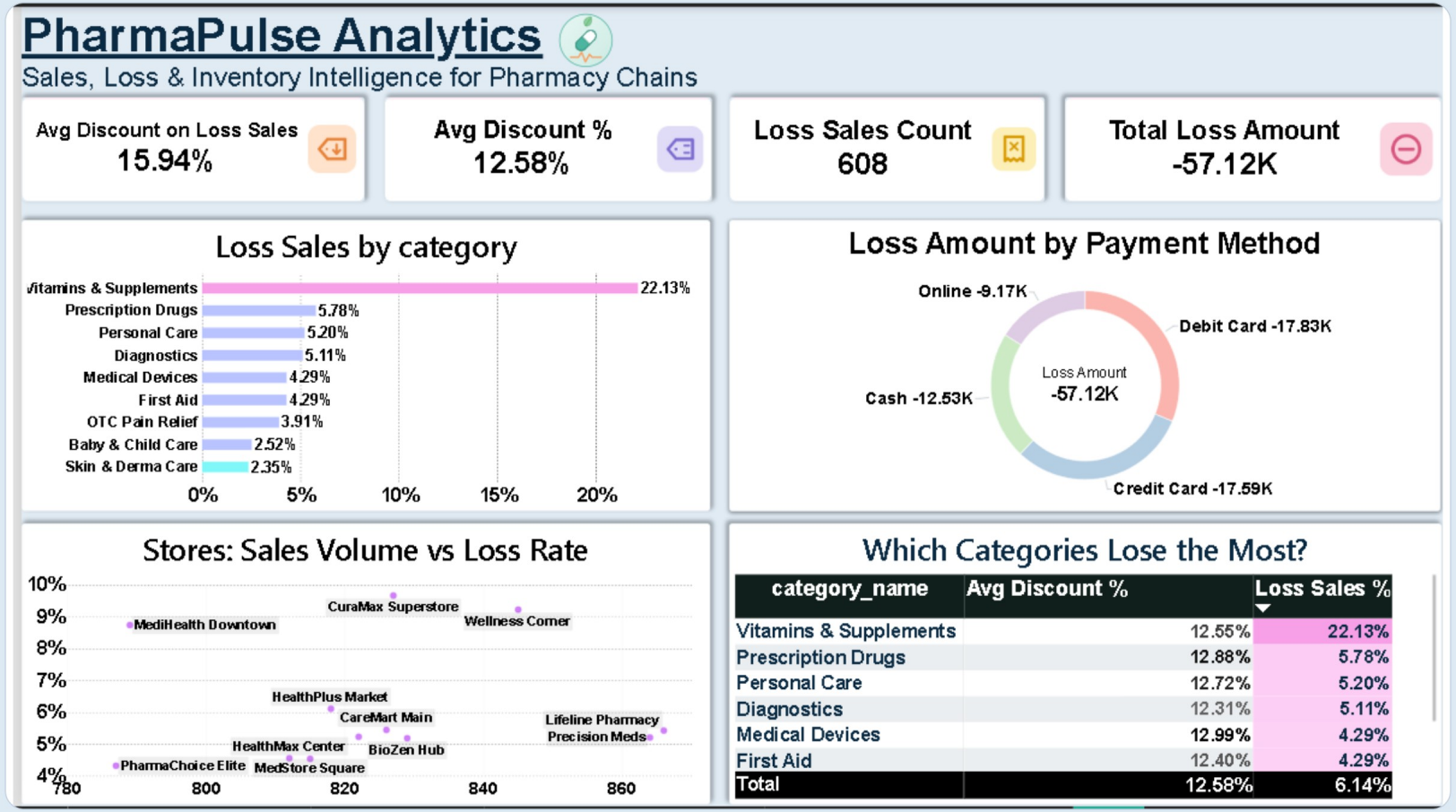
Discount	Margin	Loss rate
0-5%	40.9%	4.8%
5-10%	38.7%	5.2%
10-15%	36.5%	5.2%
15%+	21.3%	9.2%

22 queries in total: JOINs, CASE WHEN, HAVING, CTEs, LAG and RANK.

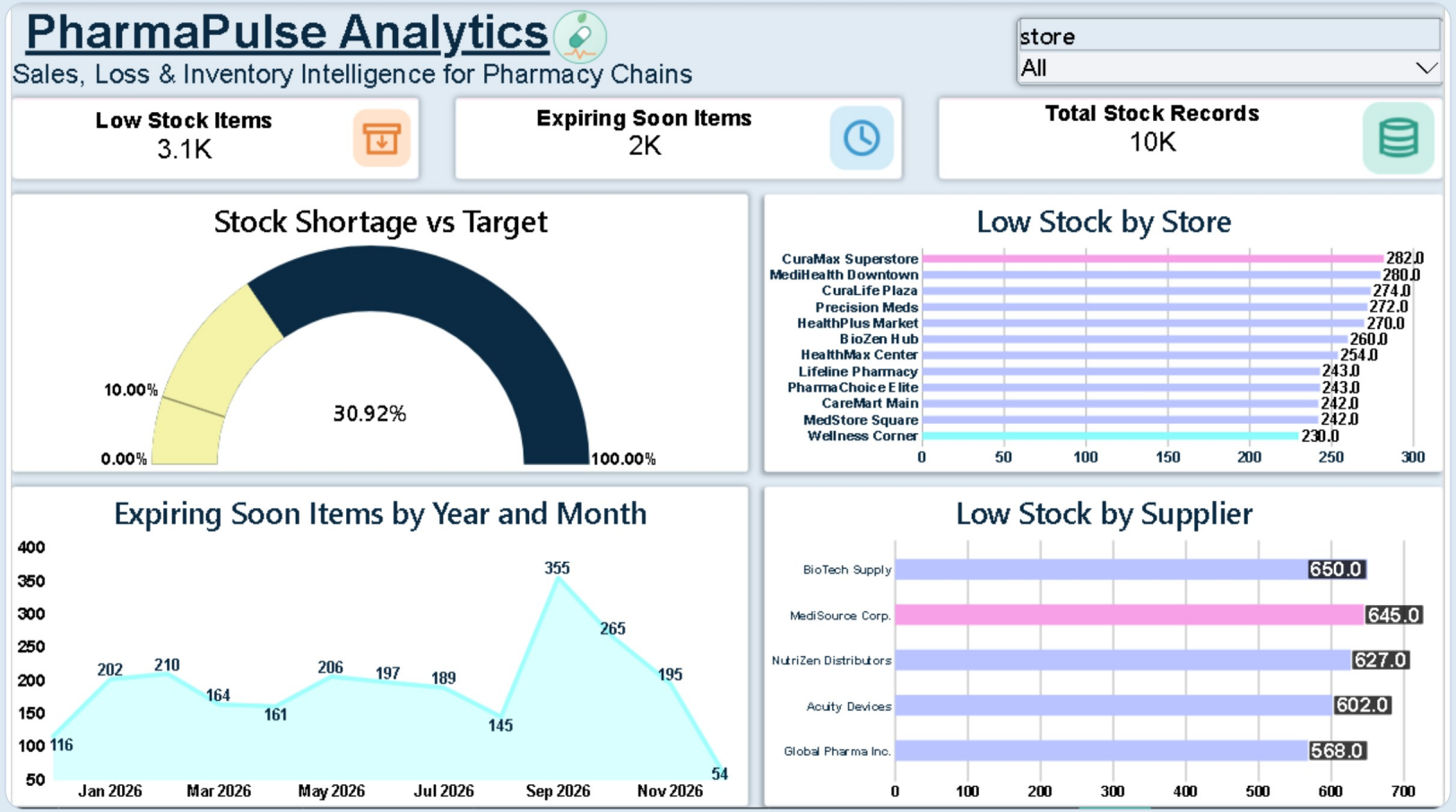
Overview: East earns the most, West has the best margin



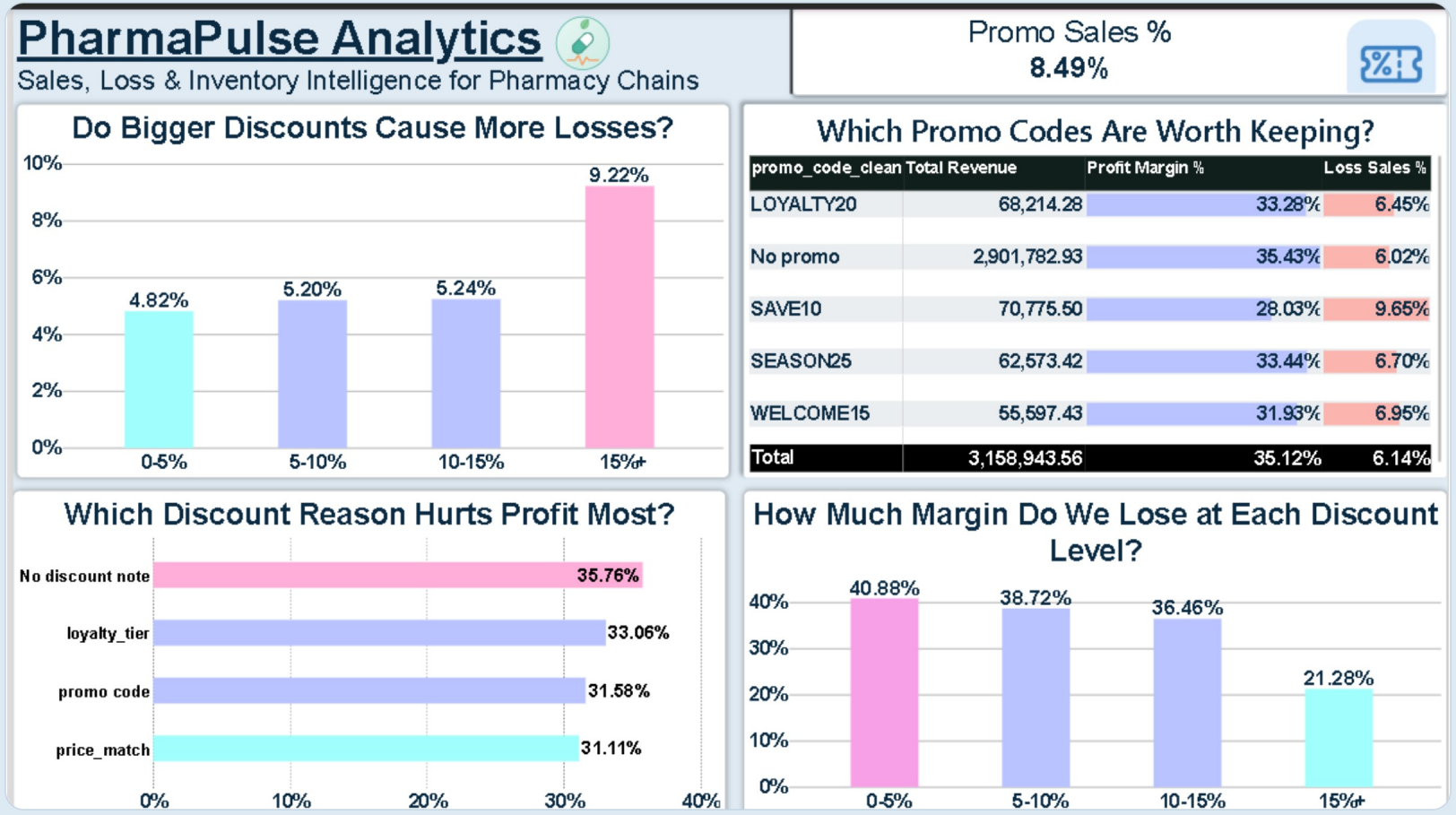
Loss: Vitamins lose 22% of their sales



Inventory: 31% of stock records are below reorder level



Discounts: above 15%, the margin is cut in half



What the data says

21.3%

margin on discounts above 15%, against 40.9% for 0-5%. Loss rate doubles to 9.2%.

22.1%

loss rate in Vitamins. 112 of its 543 products are priced below cost.

28.0%

margin on promo code SAVE10, the weakest code, with a 9.7% loss rate.

Recommendations: cap discounts near 15%, reprice the Vitamins products sold below cost, review SAVE10.

Thank you

Python · SQL Server · Power BI · DAX · Excel

Full code, queries and dashboard are on GitHub

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